

NORILSK, Russian Federation

WMO: 230780

Lat: 69.3481N

Lon: 88.2597E

Elev: 212

StdP: 14.58

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.5	-45.7	-53.2	0.2	-49.0	-49.7	0.3	-45.7	41.3	12.0	37.4	-2.8	6.5	130	0.903

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	79.6	63.8	75.0	61.3	70.2	59.3	64.7	75.8	62.8	72.8	60.7	69.2	8.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	79.9	68.7	58.5	73.6	67.1	56.3	68.0	65.1	29.9	75.6	28.5	72.2	27.0	69.1	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.9	26.8	23.3	-51.2	82.4	4.2	4.2	-54.2	85.4	-56.7	87.9	-59.0	90.3	-62.0	93.4		
(5)			-51.8	67.4	4.3	2.9	-54.9	69.5	-57.4	71.2	-59.8	72.8	-62.9	74.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	-17.3	-15.0	1.6	11.7	26.5	50.2	59.7	52.0	40.3	19.2	-7.5	-9.6		
	DBStd	29.86	18.28	17.03	17.00	13.02	9.81	10.00	8.70	7.42	7.92	12.76	14.83	17.30		
	HDD50	12325	2088	1820	1501	1149	729	121	18	65	305	956	1726	1847		
	HDD65	17275	2553	2240	1966	1599	1194	454	212	408	740	1421	2176	2312		
	CDD50	587	0	0	0	0	0	127	320	126	14	0	0	0		
	CDD65	63	0	0	0	0	0	10	49	4	0	0	0	0		
	CDH74	464	0	0	0	0	0	76	371	17	0	0	0	0		
	CDH80	78	0	0	0	0	0	11	66	1	0	0	0	0		
	WSAvg	10.9	12.3	11.6	12.7	12.4	11.2	9.9	8.4	9.0	8.9	10.7	11.5	12.6		
	PrecAvg	33.7	2.8	2.4	2.4	2.3	2.4	2.4	2.5	3.2	2.9	4.3	3.1	3.2		
	PrecMax	44.5	5.4	4.3	6.2	4.8	3.9	4.3	4.7	5.6	5.9	6.9	5.6	5.8		
(15) Precipitation	PrecMin	20.9	0.6	0.6	0.4	0.2	0.8	0.6	0.2	1.1	1.3	1.3	0.6	0.4		
	PrecStd	4.9	1.3	1.0	1.2	1.1	0.8	0.9	1.2	1.0	1.0	1.4	1.3	1.3		
	DB	20.5	23.0	34.2	39.5	54.7	81.5	85.1	76.1	63.3	48.3	27.5	26.3			
	MCWB	18.8	21.4	30.8	34.5	43.2	65.2	65.2	61.4	53.4	44.8	25.9	25.0			
	2% DB	15.0	19.5	32.2	35.4	48.0	75.6	81.7	71.0	58.7	40.2	24.3	22.6			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	13.7	18.2	29.8	32.1	40.1	60.6	63.9	59.8	51.4	37.9	23.0	21.2			
	DB	11.0	13.6	29.5	33.0	42.4	70.6	78.7	67.2	55.2	36.8	20.2	17.6			
	MCWB	9.8	12.4	27.0	30.2	37.2	58.6	63.1	57.9	49.3	34.4	18.9	16.3			
	5% DB	6.9	9.6	25.7	30.2	38.5	65.6	74.0	63.4	51.5	34.2	13.3	14.1			
	MCWB	5.6	8.7	23.9	27.6	34.7	56.0	61.6	56.5	46.8	32.1	12.1	12.8			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	18.9	21.6	31.7	34.9	44.4	67.1	68.5	63.7	55.0	44.3	26.6	25.4			
	MCDB	20.6	22.7	33.4	38.0	52.5	77.8	79.1	72.2	60.3	50.1	27.0	26.4			
	2% WB	13.9	18.0	29.6	32.6	40.5	63.4	65.5	61.5	52.4	38.1	22.9	21.2			
	MCDB	15.2	19.4	32.0	35.3	47.2	71.6	77.7	68.8	58.0	40.1	24.0	22.6			
	5% WB	9.8	12.6	27.0	30.4	37.5	60.0	64.0	59.3	49.8	34.7	18.8	16.4			
(33) Mean Daily Temperature Range	MCDB	10.9	13.6	29.2	32.8	42.5	68.1	75.4	64.8	54.3	36.3	20.1	17.4			
	10% WB	5.8	8.8	23.8	27.5	35.1	56.8	62.4	57.4	47.0	32.4	12.2	12.9			
	MCDB	6.9	9.6	25.7	30.0	38.2	65.3	73.0	62.7	51.2	34.2	13.2	14.1			
	5% DB	11.5	12.5	12.9	14.1	10.6	13.3	13.4	11.1	9.1	9.3	11.2	12.7			
	MCDBR	13.0	13.2	10.8	15.2	14.9	20.8	19.1	17.1	13.9	9.0	12.5	13.2			
(40) Clear-Sky Solar Irradiance	MCWBR	12.4	12.5	9.4	13.8	10.0	10.8	7.6	8.6	8.5	7.3	11.8	12.8			
	MCDBR	13.2	13.7	11.0	15.0	13.4	19.4	17.2	14.4	12.5	8.9	12.8	13.5			
	MCWBR	12.7	13.1	10.1	13.9	9.5	10.6	7.7	8.5	8.4	7.6	12.3	13.1			
	taub	0.228	0.262	0.320	0.347	0.386	0.397	0.426	0.417	0.344	0.265	0.215	N/A			
	taud	1.840	1.924	1.935	1.958	1.971	2.229	2.185	2.156	2.270	2.027	1.867	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	4	138	205	242	247	251	236	220	207	135	3	0			
	Edh at Noon	2	16	31	41	46	37	37	34	23	13	1	0			
(44) All-Sky Solar Radiation	RadAvg	4	114	602	1309	1853	1696	1571	980	503	163	18	0			
	RadStd	1	15	66	110	140	145	236	112	52	30	2	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page